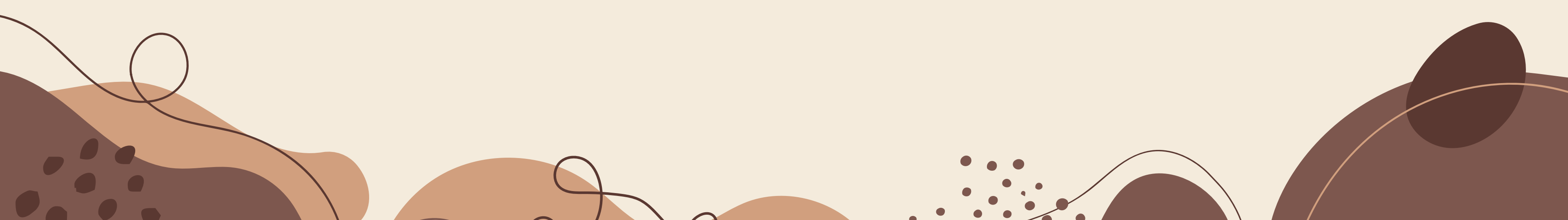




NextHikes

Project Title: Rossmann Store Sales
Forecasting
Course: Machine Learning / Data Science
Submitted by: Tanya Gera





Problem Statement

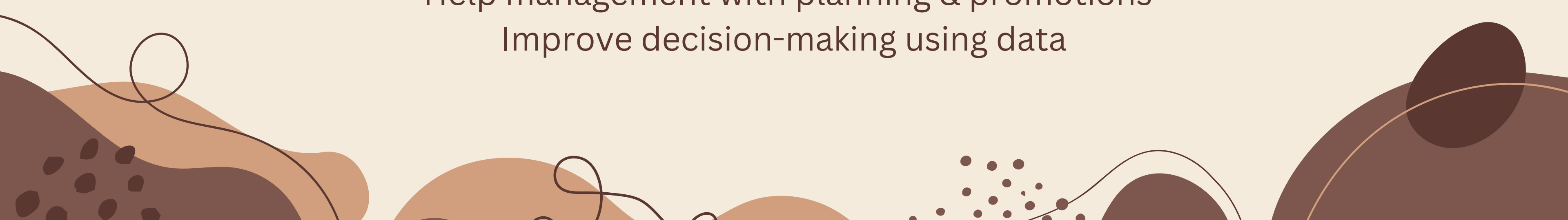
Rossmann operates over 3,000 drug stores in 7 European countries.

The goal of this project is:

To predict daily store sales

Help management with planning & promotions

Improve decision-making using data



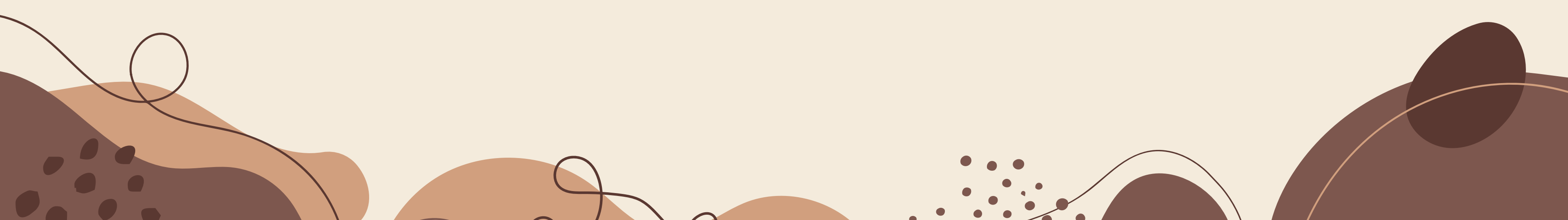


Dataset Description

- Source: Rossmann Kaggle dataset
- Includes:
 - 1M+ historical sales records (train)
 - 1,115 stores
 - Store info (competition, assortment, type)
 - Holiday & promo data
- Files used:
 - train.csv
 - test.csv
 - store.csv



Data Preprocessing

- Steps performed:
 - Handling missing values
 - Feature selection
 - Train-Test split
 - Scaling using StandardScaler
 - Time-based validation
- 

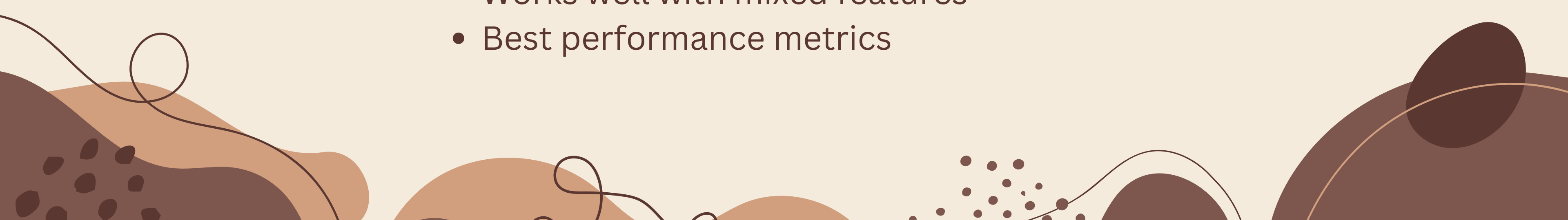


Model Selection

Tried multiple models:

- Linear Regression
- Random Forest
- XGBoost (chosen model)

Why XGBoost?

- Handles non-linear relationships
 - Works well with mixed features
 - Best performance metrics
- 

Model Performance

XGBoost Validation Performance:

- MSE: ~18020
- RMSE: ~134
- R² Score: ~0.998

XGBoost:

MSE: 18020.28992094781

RMSE: 134.239673423872

R2: 0.9981262679918751



Deep Learning Model

Model Used:

- LSTM Neural Network

Architecture:

- Input Layer
- LSTM Layer
- Dense Layer

Evaluation Metrics:

MAE: 5503

RMSE: 6666



Model comparison

Although the ML model showed very strong performance (RMSE ≈ 134), the Deep Learning model was selected for deployment to demonstrate time-series modeling capability using LSTM.

Model Serialization

To deploy the model:

DL model saved as .h5

Scaler saved using pickle

This allows the model to be reused without retraining.

Deployment

Deployment was implemented using Streamlit.
The application allows users to input store features and obtain real-time sales predictions.

Rossmann Store Sales Prediction

Enter Store Features Below:

Store ID

1

-

+

Day of Week

1

Promo

0

▼

School Holiday

0

▼

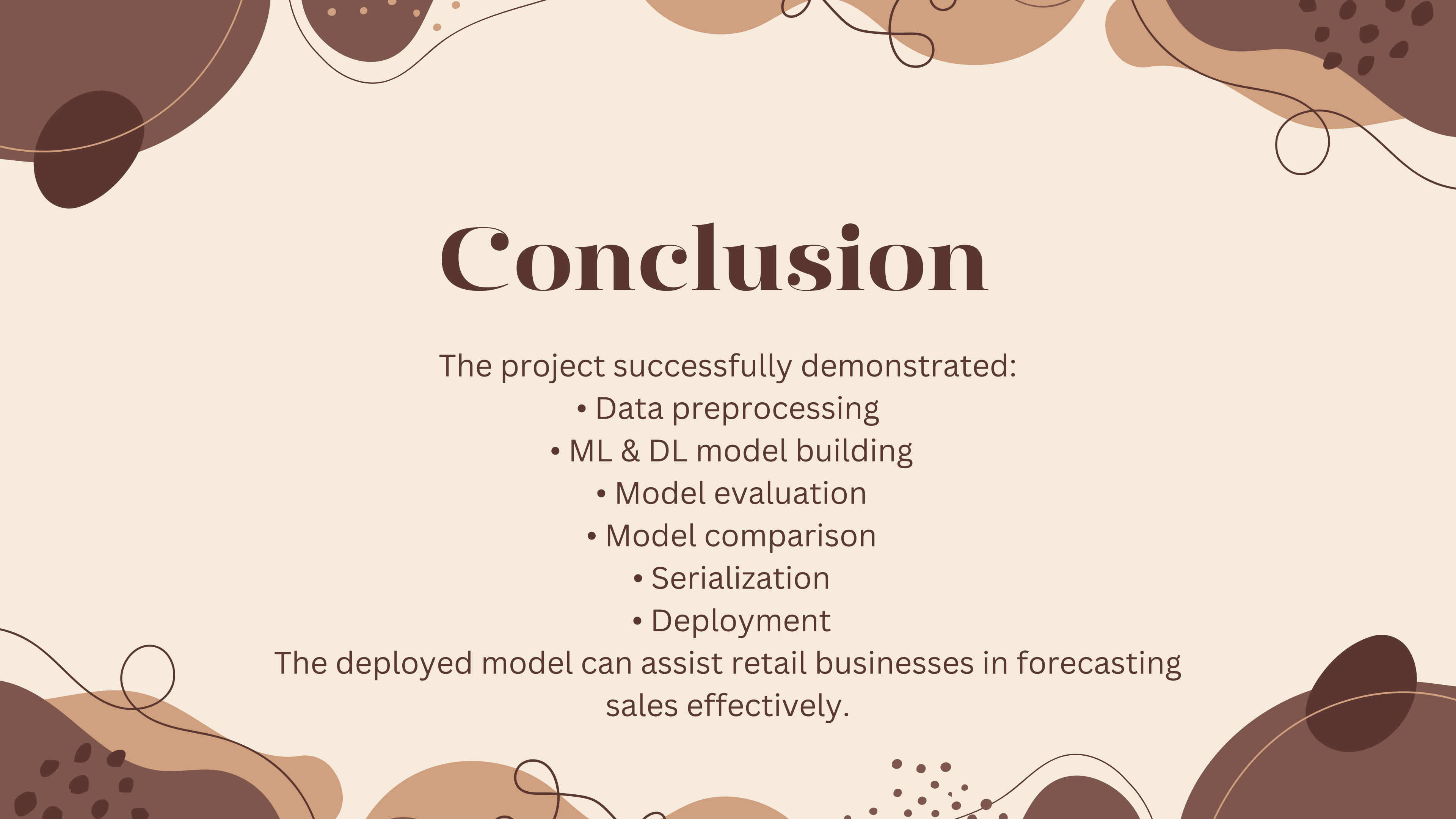
Customers

0

-

+

Predict Sales



Conclusion

The project successfully demonstrated:

- Data preprocessing
- ML & DL model building
 - Model evaluation
 - Model comparison
 - Serialization
 - Deployment

The deployed model can assist retail businesses in forecasting sales effectively.

The background features abstract, organic shapes in various shades of brown and tan. These shapes are scattered across the top and bottom edges, with some containing small dots or swirls. Thin, dark brown lines meander across the composition, adding a sense of movement and elegance. The central text is a large, dark brown serif font, standing out against the light tan background.

Thank
You